SANTOSH KUMAR
Embedded Sr. Engineer

tecsantoshkumar63@gmail.com
+91| 6306753746

Versatile Embedded Systems Engineer with 4.10 years of experience in developing memory efficient firmware. Expertise in micro-controller (MCU) programming and VTS Devices with encryption. Skilled in RTOS, IOT, Secure FOTA/COTA.

TECHNICAL SKILLS

Micro-Controller	STM32, HT8/32, TI, CH32, ESP, Microchip, Raspberry Pi.
Communication Protocols & Peripherals	UART, SPI, I2C, ADC, TIMER, RTC, FDCAN, MODBUS, ETH, USB, MEMS, SPI flash, EEPROM, Serial wire
Internet Protocols	Bluetooth, Wi-Fi, TCP/IP, HTTPS, MQTTS, SMTP, SFTP, Web Socket, IRNSS, GSM, GPS, LWM2M
Programming Language	C/C++, Python
RTOS	Task Scheduling, Mutex, semaphore, Event flags, Memory Management
Tools	STM32CubeIDE, IAR-IDE, KILL-IDE, ESP-IDF, HT32-IDE, MPLAB IDE, ICP IDE, Git, Visual Studio, VS Code, Hercules,
Cryptography	AES 128/256 (ECB, CBC), OpenSSL, WolfSSL, MbedTLS
Hardware Design	Circuit Designing using Altium Designer
Sensor Interface	IR, Ultrasonic, RFID, DHT11, OHRL, GWL, MEMS
Display Interface	Segmented display, Matrix display,GLCD, TFT display

WORK HISTORY

28 April 2025 - Currently Working

❖ **IoTfy Solutions Private Limited – New Delhi**

- Leading firmware design and validation for Inergy BLDC motor driver and exhaust fan control systems, including single-phase and 3-phase HV/LV motor platforms.
- Developed and optimized embedded control logic for Havells BLDC and combined chimney modules, integrating RTC, Temp SNS, and heater control over I²C and ESP32C3.
- Designed and implemented touch display interfaces and Wi-Fi–enabled control systems for Glen smart geysers and chimneys using WT32-SC01 and ESP-IDF frameworks.
- Delivered complete IoT chimney and smart ventilation solutions, integrating TFT touch displays (1.28”, 3.5”, 4.0”), IR control, and mobile app connectivity.
- Debugged and validated PICL exhaust fan and ventilation PCBs, ensuring reliable BLDC motor performance and control logic stability.
- Collaborated with multi-brand validation and R&D teams for cross-product integration, firmware optimization, and client demonstration builds.

12 December 2022 - 10 April 2025

❖ **EDS INDIA Pvt Ltd- New Delhi**

- 4G Cluster: TCP/IP, CAN, GNSS, Motor Control using STM3 & QUCTEL 4G Modem.
- Pump Monitoring System: MQTT, HTTP, Modbus (RS-485), BLE, Flash, using STM32 & QUCTEL 4G Modem.
- DMS with Vice Box: 4G Gateway, UART, Modbus (RS-485), Flash, using STM32 & QUCTEL 4G Modem.
- JIO VTS: AIS-140, TCP, MQTT, HTTP, LWM2M, AES 128/256, CAN, IRNSS, Bootloader, FOTA, using STM32F0 & QUCTEL EC200U
- TML VTS: AIS-123, MQTT, HTTP, AES128, FDCAN, IRNSS, Secure Bootloader, FOTA, using STM32 & QUCTEL 4G Modem.
- Accolade VTS: AIS-140, TCP, MQTT, HTTP, CDAC, FDCAN, IRNSS, Bootloader, FOTA, using STM32 & QUCTEL 4G Modem

17 August 2021 – 30 November 2022

❖ **Su-Vastika System Pvt Ltd - Gurugram, Hariyana**

- Driver Development for High-performance Computing Controllers STM8, STM32, DSPIC30F3011
- Inverter Software and Inverter Based LCD Software using STM8, DSPIC30f3011
- Using ESP32 hardware and ESP-IDF software, I designed the entire architecture for a system that provides remote monitoring and control of UPS units, capable of supporting the installation of over 1,000 devices.

PERSONAL DETAILS

Father's Name	Mr. Balchndra
Date of Birth	25/03/1998
Marital Status	Single
Permanent Address	Byoti Vijayeepur Khaga Fatehpur (u.p.)
Present Address	Janakpuri, Mayapuri, Dehli

EDUCATION

2017 - 2021	B. Tech Completed in Electrical & Electronics branch from SR Group of Institutions & Management Technology BKT. Lucknow
2015 - 2017	10th and 12th Completed in Ram Gopal Tripathi Inter College Vijayeepur Khaga Fatehpur

DECLARATION

I Confirm that the information provided by me is true to the best of my knowledge and belief.

Date: 09/08/2026

Sd/-
Santosh Kumar